Identifying Spam Tests Through Machine Learning

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Introduction

Your phone dings. You check your texts, expecting to see a message from a friend or family member, but instead, you see an ad. Spam text messages remain a common annoyance. In one survey, 29% of respondents reported receiving spam messages daily or multiple times a day, with another 28% reporting receiving them weekly (Crestodina 2025). Spam text messages can even be harmful. Sixty-one percent of U.S. adults reported receiving spam text messages at least weekly that attempted to get the receiver's personal information (Gottfried, Park, and Anderson 2025).

The SMS Spam Collection dataset (Almeida and María Gómez Hidalgo 2011) offers a set of spam text messages and genuine text messages ("ham") that can be used to build and test models to identify spam messages. The dataset consists of two variables: the classification ("spam" or "ham") and the text of the message. The dataset contains a total of 5,574 messages, 747 of which are spam and 4,827 of which are ham (Almeida, María Gómez Hidalgo, and Yamakami 2011).

One important consideration when comparing models is the cost of a false positive (identifying a legitimate message as spam) versus the cost of a false negative (identifying an unwanted message as ham). Text messages can be used for time-sensitive communication, with the expectation that the receiver will receive an alert immediately. This can include flight changes or fraud alerts (CTIA 2023) or emergency notifications, such as an active tornado warning. Missing an urgent, or less time-sensitive but still meaningful message, can cause harm given how often people rely on text messaging for communication. Americans exchanged nearly 2.2 trillion text messages in 2025 (CTIA 2025a).

However, failing to identify a spam message can be harmful if the message is one of the many that attempts to defraud the receiver or trick them into handing over personal information. While ads are still the most common form of spam text messages, 37% of survey respondents said that the spam text messages they received asked for personal information (Crestodina 2025). On the other hand, while the risk is real, seventy-one of U.S. adults feel they know at least a fair amount about how to avoid falling for an online scam (a category that includes spam text messages) (Gottfried, Park, and Anderson 2025). The Federal Trade Commission, among other organizations, provides guidance on how to recognize and report text message scams (Federal Trade Commission 2022), which may help decrease the likelihood that someone seeing a spam message is actively harmed.

Considering the trade-offs, I determined that recall should be weighted more heavily than precision–it was more important that someone not miss a real text message than receive a spam message. Therefore, when comparing models, I calculated the F value, which combines precision and recall, with a beta of 0.75, weighting precision 0.75 times that of recall (Irizarry, n.d.a; "What Is F-Beta Score?" 2025).

After evaluating multiple models separately and as ensembles, I found that the model using a random forest best identified text messages as spam or ham.

Methods and Analysis

Data Cleaning

The SMSSpamCollection file is a tab-separated text file, with one line per message. I parsed it into rows and columns using `read_tsv` to create two columns, the message type ("spam" or "ham") and the text of the message.

Table 1: Example messages from the dataset

type	message
ham	Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got amore wat...
ham	Ok lar... Joking wif u oni...
spam	Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entry question(std txt rate)T&C's apply 08452810075over18's
ham	U dun say so early hor... U c already then say...
ham	Nah I don't think he goes to usf, he lives around here though
spam	FreeMsg Hey there darling it's been 3 week's now and no word back! I'd like some fun you up for it still? Tb ok! XxX std chgs to send, £1.50 to rcv
ham	Even my brother is not like to speak with me. They treat me like aids patent.
ham	As per your request 'Melle Melle (Oru Minnaminunginte Nurungu Vettam)' has been set as your callertune for all Callers. Press *9 to copy your friends Callertune
spam	WINNER!! As a valued network customer you have been selected to receivea £900 prize reward! To claim call 09061701461. Claim code KL341. Valid 12 hours only.
spam	Had your mobile 11 months or more? U R entitled to Update to the latest colour mobiles with camera for Free! Call The Mobile Update Co FREE on 08002986030

When I first looked at the most popular words across all messages, I saw "gt" and "lt" in the list. While "2", "ur", and "4" are all commonly-known texting abbreviations, "gt" and "lt" seemed like odd words to be sent so frequently.

```
messages |>
  unnest_tokens(word, message, to_lower=TRUE, strip_punct=TRUE) |>
  filter(!word %in% stop_words$word) |>
  count(word) |>
  top_n(10, n) |>
  mutate(word = reorder(word, n)) |>
  arrange(desc(n)) |>
  kable(caption="Most popular words across all messages")
```

Table 2: Most popular words across all messages

word	n
call	588
2	513
ur	391
gt	318
4	316
lt	316
free	283
day	221
time	218

word	n
love	207

Reviewing the messages that contain 'lt' or 'gt' indicated that these were HTML-encoded entities. Using the `textutils` package, I decoded any encoded HTML entities and stripped any HTML tags that might still have been included.

Now the most popular words all make sense as words that might be used in texts.

```
messages <- messages |> mutate(message=HTMLrm(HTMLdecode(message)))
messages |>
  unnest_tokens(word, message, to_lower=TRUE, strip_punct=TRUE) |>
  filter(!word %in% stop_words$word) |>
  count(word) |>
  top_n(10, n) |>
  mutate(word = reorder(word, n)) |>
  arrange(desc(n)) |>
  kable(caption="Most popular words after cleaning")
```

Table 3: Most popular words after cleaning

word	n
call	588
2	513
ur	391
4	316
free	283
day	221
time	211
love	207
send	196
text	193

Next I standardized phone numbers. Even in the small sample of texts reviewed, I could see at least one phone number, and I knew that spam messages might instruct someone to call or a text a certain number. The presence of "call" and "text" as commonly used words supported that. But since phone numbers can take a variety of forms, I standardized several known forms to be all digits. This would also help me later when testing whether the presence of long strings of numbers, such as phone numbers or account numbers, would be indicative of a spam message.

```
convert_phone_numbers <- function(s) {
  s |>
    # Handle formats 000-000-0000, 000.000.000, and 000 000 0000
    str_replace_all("(\\d{3})[-,\\.\\s](\\d{3})[-,\\.\\s](\\d{4})",
                    "\\1\\2\\3") |>
    # Handle format (000) 000-0000
    str_replace_all("\\((\\d{3})\\)\\s?(\\d{3})-(\\d{4})", "\\1\\2\\3")
}
messages <- messages |>
  mutate(message = convert_phone_numbers(message))
```

Table 4: Most commonly used words in ham and spam messages

word	n	word	n
2	266	call	287
ur	202	free	180
call	183	2	141
day	167	txt	119
love	166	mobile	110
time	161	text	109
4	154	ur	109
home	130	stop	98
ü	129	4	97
lor	128	claim	94

Analysis

Before doing any analysis, I split the data into training and test sets, so that I could look for differences between ham and spam text messages in the training set only. Knowing that, of course, the data set represents a small fraction of the spam and ham text messages sent, I wanted to use the test set as a stand in for the unknown or yet-to-be-set text messages that I couldn't analyze.

Given the overall small size of the data set, I wanted to use as much data as possible for analysis and training. However, the test set still needs to be large enough to contain meaningful data. A split of 10%-20% of the data for the test set is common (Irizarry, n.d.b). I chose 20%, so there would be a little over one thousand messages in the test set. As 2024 study proposed a minimum data set sizes of N=500-1000 to reduce overfitting (Zantvoort et al. 2024), the remaining training set size of around 4,400 messages should be sufficient.

Most Commonly Used Words I compared the mostly commonly-used words in the training set for ham messages versus spam messages.

There are some apparent differences. "Free" is the second-most-often used words in spam texts but does not appear in the top 10 at all for ham texts. However, there is also overlap. "Call", "ur," "2", and "4" appear in both lists. Therefore, merely identifying words that are used often in spam messages are likely to result in a lot of false positives. Instead, I calculated the odds ratio for each word to identify those that were most likely to be in spam messages. (Irizarry, n.d.a)

word	ham	spam	odds_ratio
claim	0	94	441.55
prize	0	72	337.94
won	0	56	262.90
150p	0	55	258.21
tone	0	45	211.46
guaranteed	0	40	188.12
18	0	37	174.13
cs	0	36	169.46
1000	0	33	155.49
1.50	0	32	150.83
500	0	32	150.83
awarded	0	29	136.86
150ppm	0	27	127.55
ringtone	0	27	127.55

word	ham	spam	odds_ratio
5000	0	23	108.95

This demonstrated that were definitely words that appeared more often in spam messages than in ham messages. There were 7,513 unique words used in the training set. 2,087 of those words had an odds ratio greater than 1, indicating a positive correlation between the presence of that word in the message and the message being spam. 759 of the words had an odds ratio greater than 10, indicating an even stronger positive correlation. (Bobbitt 2021)

I then counted the number of times a word in the list of top spam words (those that are at least 10 times more likely to appear in a spam message than a ham message in the training set) and added that data to both the training and test sets.

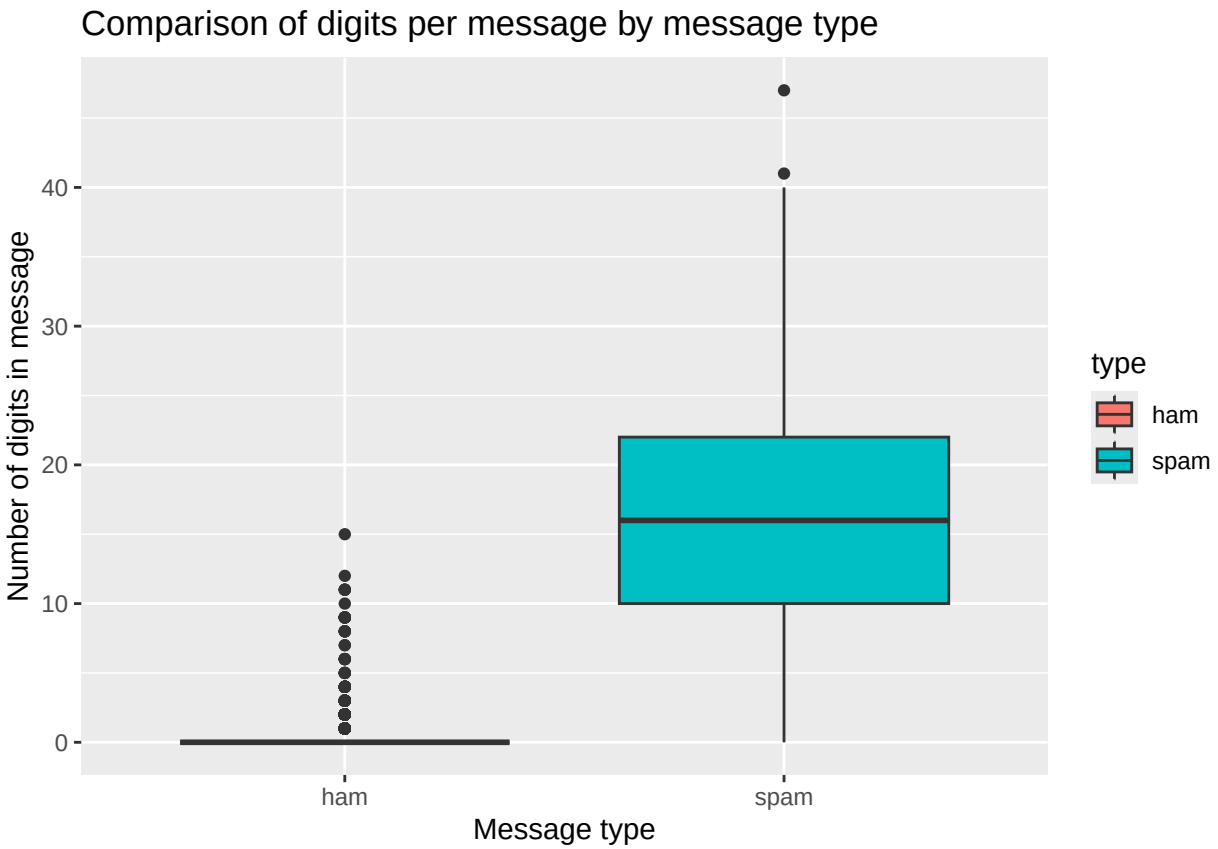
```
# Get the words with an odds ratio of 10 or more. That's the list of words
# that are at least 10x more likely to be in a spam message than a ham message.
top_spam_words <- ham_spam_odds |> filter(odds_ratio >= 10) |> pull('word')

# Count how many times any of those words appear in the message
count_spam_words <- function(data, word_list = top_spam_words) {
  data |>
    mutate(top_spam_words_count = sapply(message, function(m) {
      words <- list_c(tokenize_words(m, lowercase = TRUE, strip_punct=TRUE))
      sum(words %in% top_spam_words)
    }))
}

train_set <- count_spam_words(train_set)
test_set <- count_spam_words(test_set)
```

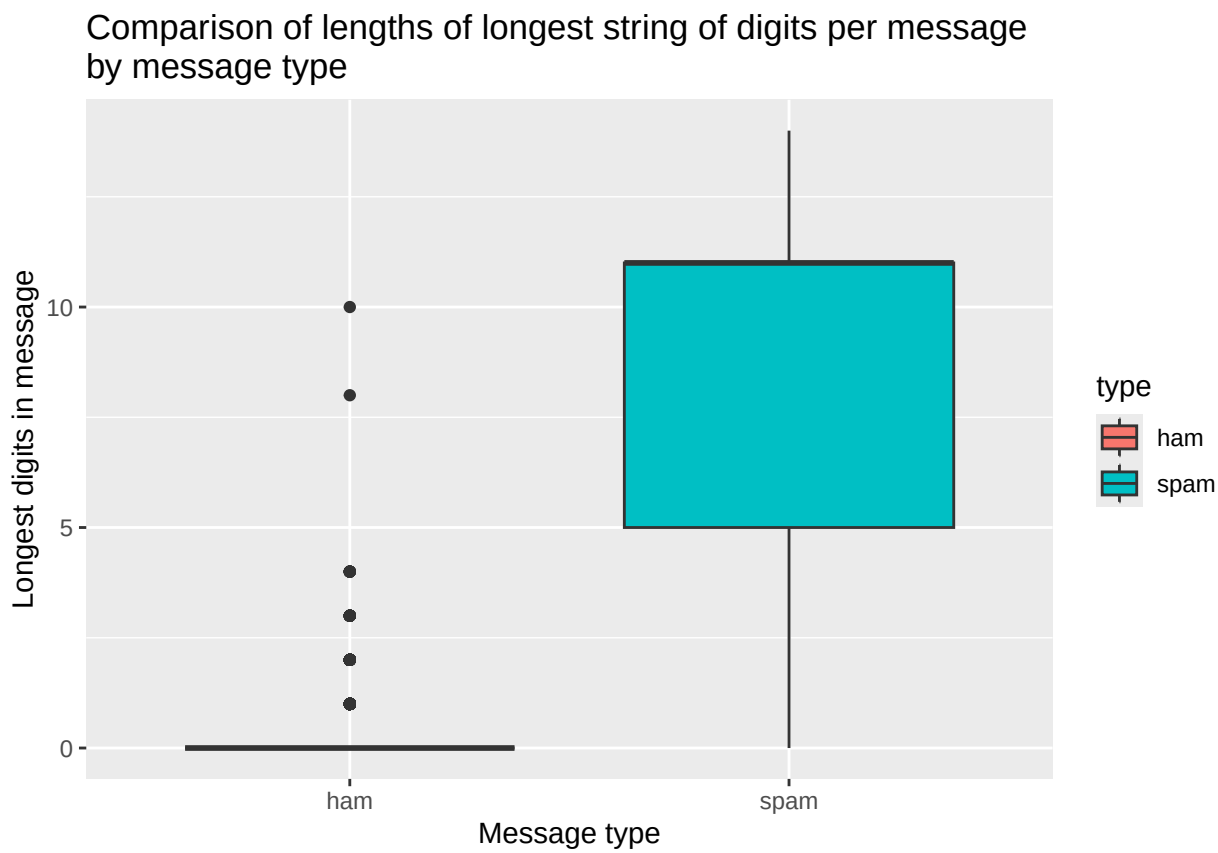
Numbers Previously I'd hypothesized that spam messages might be more likely to contain phone numbers or other long number strings, such as a (faux) bank account or package tracking number. In addition, the list of words more likely to be in spam messages contained several numbers.

First, I added two counts to my data sets: the total number of digits appearing in the message and the length of the longest unbroken string of numbers.

```
add_number_columns <- function(data) {
  data |>
    mutate(message = convert_phone_numbers(message)) |>
    mutate(num_count = str_count(message, "\\d")) |>
    mutate(longest_number = unlist(lapply(
      str_match_all(message, "\\d+"),
      function(m) {
        if (length(m) == 0) {
          0
        } else {
          max(nchar(m[,1]))
        }
      })
    ))
}
train_set <- add_number_columns(train_set)
test_set <- add_number_columns(test_set)
```

Spam messages tended to have both more digits overall and a longer string of digits than ham messages did.


Comparison of digits per message by message type
Number of digits in message
40
30
20
10
0
ham
spam
Message type
type
ham
spam



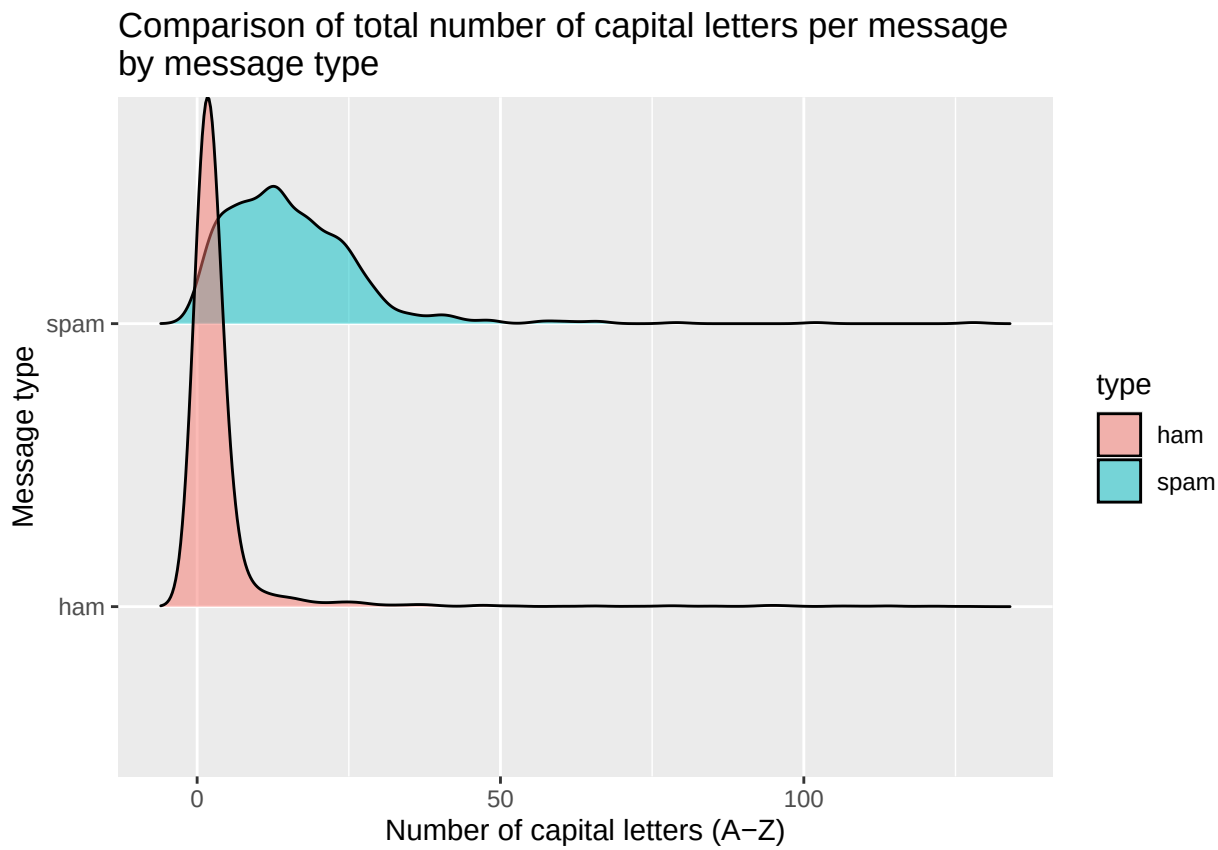
Capital Letters Like numbers, it's possible that capital letters could differ between spam and ham messages. However, with few options to convey tone over text messages (the dataset was published in 2011, which predates the global rise in emoji popularity from 2014 onward (Broni 2025)), capital letters or entire words or messages in all caps might be used by both spam messages and ham messages to convey excitement or urgency.

To see if there was any difference, I first counted both the total number of capital letters and the longest string that did not contain lowercase letters or digits. I did allow for white space and punctuation in order to detect the difference between a single word in all-caps and an entire message in all-caps, so long as the string started and ended with a capital letter (so removing trailing white space and punctuation).

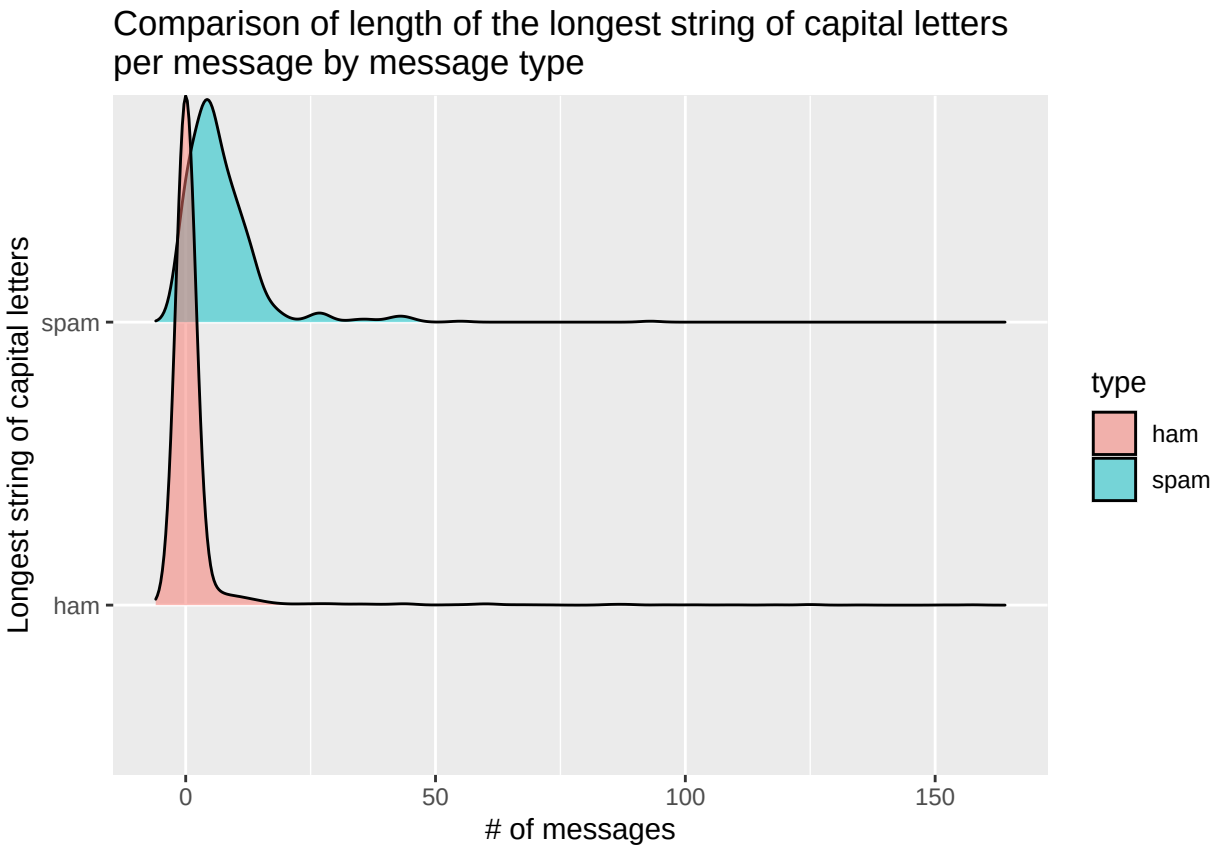
```
add_capital_columns <- function(data) {
  data |>
    mutate(capital_count = str_count(message, "[A-Z]")) |>
    mutate(longest_capitals = unlist(
      lapply(
        str_match_all(
          message,
          "[A-Z][^a-z\\d]*[A-Z]"),
        function(m) {
          if (length(m) == 0) {
            0
          } else {
            max(nchar(m[, 1]))
          }
        })
```

```
      )
    )
}

train_set <- add_capital_columns(train_set)
test_set <- add_capital_columns(test_set)
```

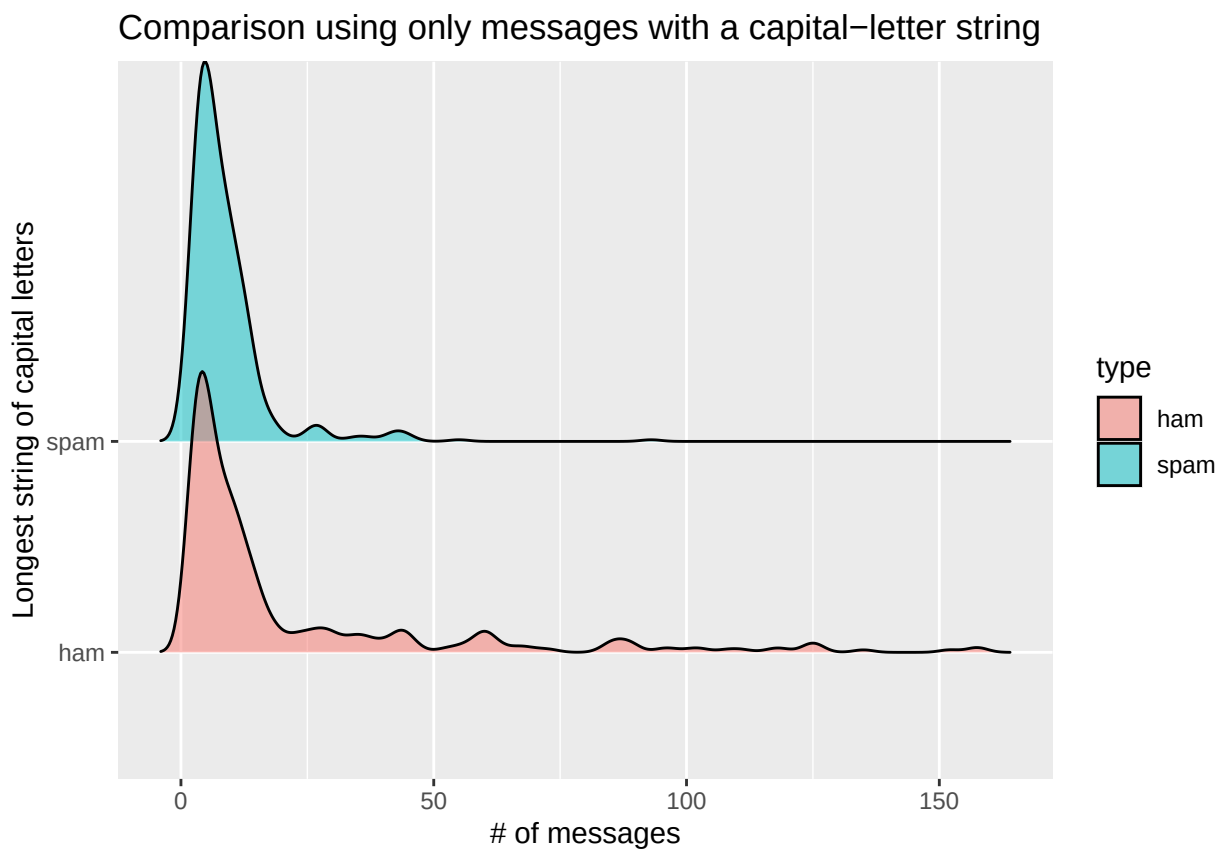
Using a ridge plot, I could see that while most ham messages had only a few capital letters, there was a long a long tail. There wass also overlap between spam and ham messages at the lower end of the range. Messages with 5 to 30 capitals were more likely to be spam messages, but then spam messages also had a long tail.



Comparing the length of the longest string of capital letters, I saw a similar pattern. Both ham and spam messages had long tails and both overlapped at the bottom of the range, with no string of two or more capital letters. However, while ham messages were much more likely to have no string than to have any length of capital letters in a row, spam messages had a wider distribution with a peak around a string of 5-10 characters.



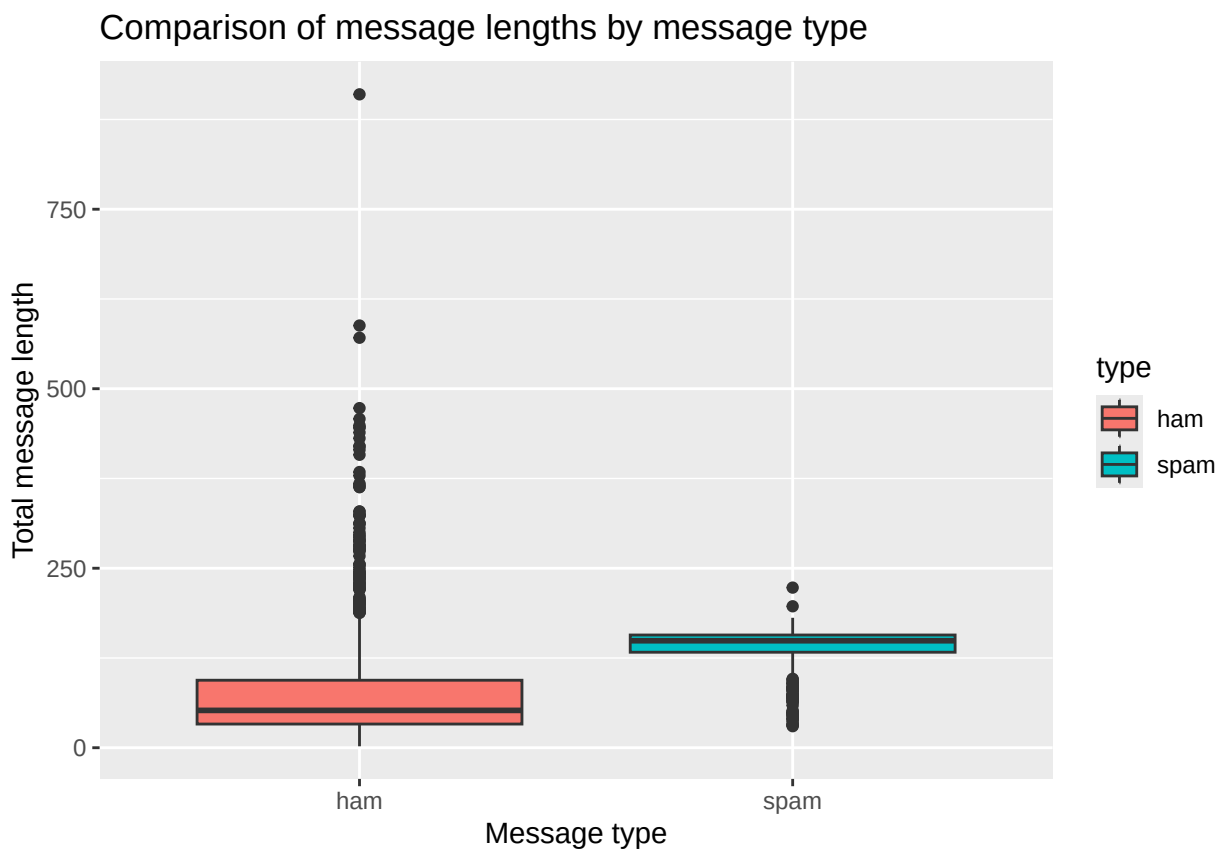
Looking only at messages where `longest_capitals` is greater than 0, I could see that most of the difference between spam and ham messages was driven by many ham messages not having a multi-capital-letter string at all. With those removed, it was more apparent that while spam messages peaked at a slightly greater number, ham messages had a more substantial tail. This suggests that while having a `longest_capitals` length of 0 might give credence to a message being ham, the length of the string beyond that doesn't seem to be a clear indicator one way or another.



Message length Finally, I counted the total character length of the messages to examine if there were any differences.

```
train_set <- train_set |> mutate(msg_length = nchar(message))
test_set <- test_set |> mutate(msg_length = nchar(message))
```

I could see that, on average, ham messages are shorter than spam messages. However, once again, both types contained a number of outliers, with messages longer than 250 characters being entirely ham messages.



Model Development

To compare model performance, I wanted to consider both precision and recall, but, as discussed earlier, giving more weight to recall. I tested each model against my previously held-out test set and compared the F-value, which computes the harmonic average of precision and recall. I used a weighted harmonic average with a beta value of 0.75, so that precision was weighted 0.75 times that of recall.

Model 1: Guessing To get a baseline, I first tested an extremely simple model, guessing whether a message is spam or ham, assuming the same proportions of spam messages that are in the training set.

```
p_spam <- sum(train_set$type == 'spam') / nrow(train_set)
guess_y_hat <- sample(
  c('spam', 'ham'),
  nrow(test_set),
  replace = TRUE,
  prob = c(p_spam, 1 - p_spam)
) |>
  factor(levels = levels(test_set$type))
guess_f_meas <- F_meas(
  guess_y_hat,
  reference = test_set$type,
  relevant = "spam",
  beta = 0.75
)
guess_f_meas
```

```
## [1] 0.1625065
```

Model 2: Using longest number As seen in the the initial analysis, messages with long strings of digits were more likely to be spam, with less overlap between the groups than the other potential indicators. For the second model, messages with the longest string of digits above a certain cutoff are classified as spam, all others as ham. To find the cutoff, I compared the F-value against the training set for cutoffs ranging from 0 to 15. A cutoff of 4 provided the optimal value, so I then tested on the test set, using that cutoff.

```
# Determined through tuning on the training set
cutoff <- 4
long_number_y_hat <- ifelse(test_set$longest_number >= cutoff, "spam", "ham") |>
  factor(levels = levels(test_set$type))
long_number_f_meas <- F_meas(
  long_number_y_hat,
  reference = test_set$type,
  relevant = "spam",
  beta = 0.75
)
long_number_f_meas
```

```
## [1] 0.9226714
```

This improves performance dramatically over guessing.

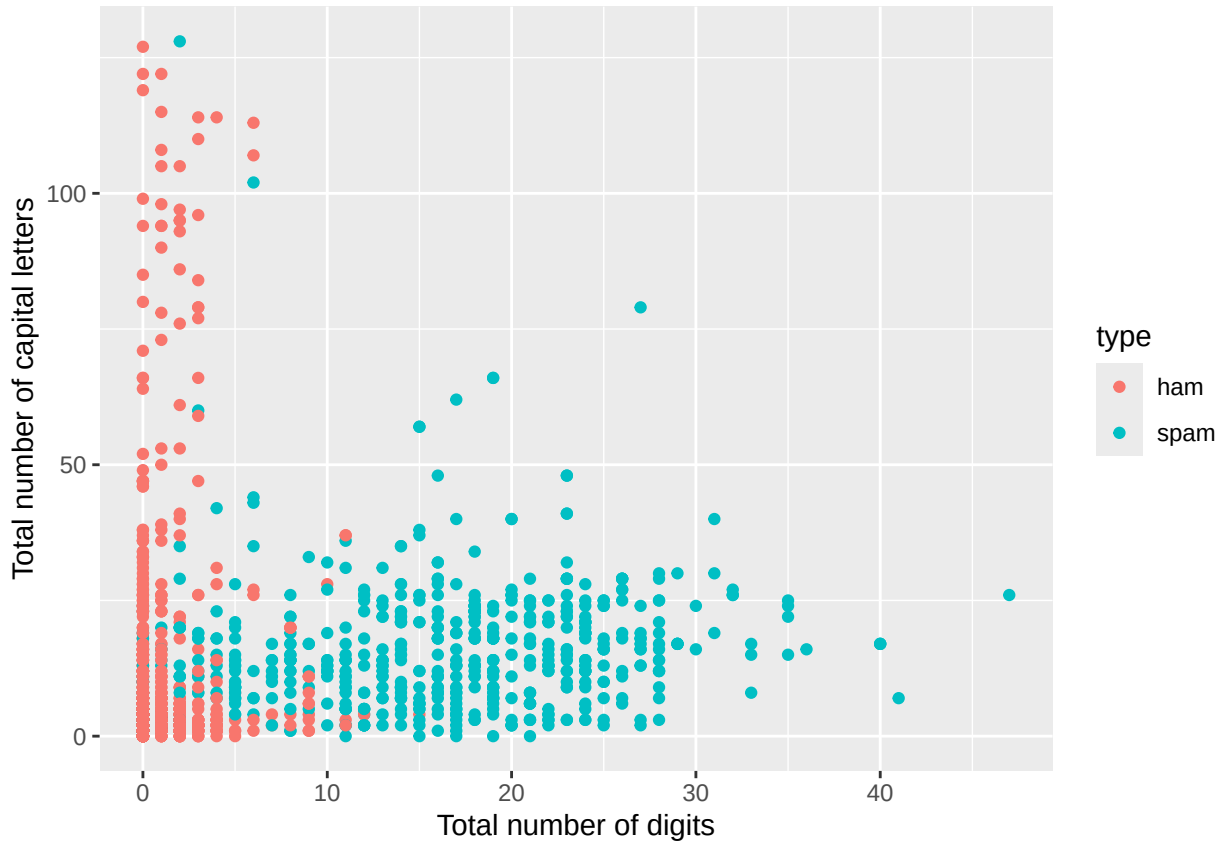
Model 3: Generalized linear model Because messages are either spam or ham, which can be represented as 0 and 1, a generalized linear model or logistic regression model makes more sense than a linear regression model. A linear regression model would allow for values below 0 or above 1. A logistic regression model converts the probability to log odds, calculating how much more likely it is that a message will be spam than that it will not be spam (Irizarry, n.d.a). Rather than a simple binary cutoff, I can use a linear regression model to calculate whether a message is likely spam or ham given a value.

```
glm_digit_length_fit <- train_set |>
  mutate(y = as.numeric(type == "spam")) |>
  glm(y ~ longest_number, data = _, family = "binomial")
# Predict the results on the test set, using the model created on the training set
# Returns the likelihood that a message is spam
glm_digit_length_p_hat <- predict(glm_digit_length_fit, newdata = test_set, type =
                                    "response")
# Translate those probabilities to predictions by saying that any message with
# a greater than 50% chance of being spam is spam
glm_digit_length_y_hat <- ifelse(glm_digit_length_p_hat > 0.5, "spam", "ham") |>
  factor(levels = levels(test_set$type))
glm_digit_length_f_meas <- F_meas(
  glm_digit_length_y_hat,
  reference = test_set$type,
  relevant = "spam",
  beta = 0.75
)
glm_digit_length_f_meas
```

```
## [1] 0.8972943
```

This did less well than finding a cut-off. While messages with longer numbers are more likely to be spam, the data is not approximately bivariate normal.

I also considered looking at more than one variable. For example, messages with more numbers and with more capital letters both tended to be spam messages.



However, I could see that while the data is mostly grouped between ham and spam messages, the relationship is not linear.

Model 4: K-Nearest Neighbor K-Nearest Neighbor (KNN) identifies the k closest data points to an item and makes the prediction based on the average of the closest data points. One advantage it has over logistic or linear regression here is that it does not assume the data is distributed in any particular way. Since ham and spam messages tend to be clustered near similar messages, predicting the classification based on the nearest other messages might yield better results. ("K-Nearest Neighbor(KNN) Algorithm" 2026)

I used the `caret` package to train the model and perform cross-validation with bootstrap samples to determine which value of k performed best. Bootstrap sampling takes a sample with replacement of the training dataset to approximate the full dataset. The `train` method does this 25 times (the default value) for each parameter and calculates the accuracy. This approximates the distribution across the full dataset. I used the same weighted F value for accuracy as I had been using to compare models. Whatever parameter value gives the highest accuracy when the 25 tests are averaged together is what's used in the final model when the outcome is used with `predict`.

```
knn_fit <- train(
  type ~ .,
  method = "knn",
  data = train_set_features,
  trControl = control,
  metric = "F1",
  tuneGrid = data.frame(k = c(3, 5, 7, 9, 11))
)
knn_fit$bestTune
```

```
##   k
## 3 7
```

The model gave the best performance when predicting using the 7 nearest neighbors. I then tested the final model against the test set.

```
knn_y_hat <- predict(knn_fit, newdata = test_set, type="raw")
knn_f_meas <- F_meas(
  knn_y_hat,
  reference = test_set$type,
  relevant = "spam",
  beta = 0.75
)
knn_f_meas
```

```
## [1] 0.9226714
```

Model 5: Classification Tree Another way to classify messages is to look at various combinations of factors. For instance, I had already seen that messages with long strings of numbers were likely to be spam. Messages with more than 5 capital letters in a row were also likely spam unless had have more than 50 capital letters, in which case, they were likely ham. Messages that contained 1 or more words in the list of words more likely found in spam messages were likely to be spamm and so on.

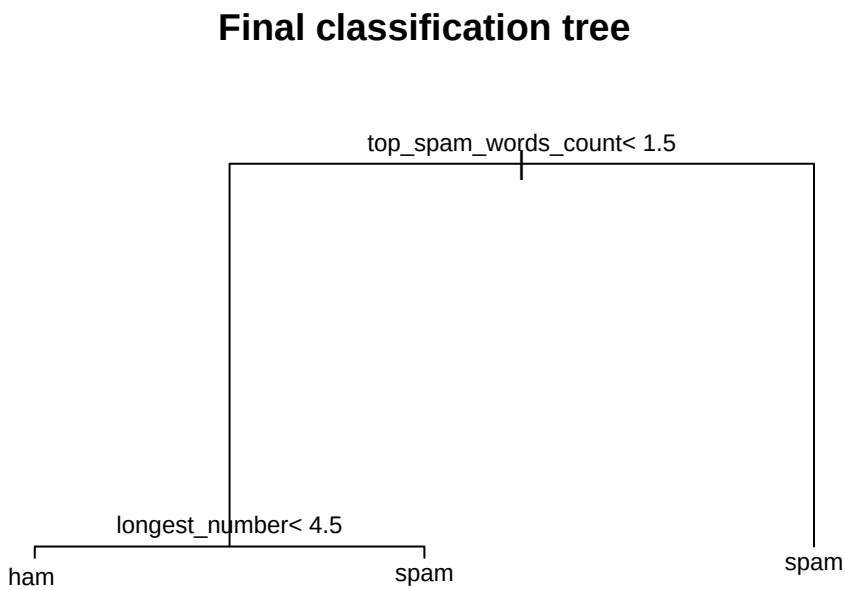
A classification tree establishes a set of rules or series of if/else conditions to determine the prediction. The complexity parameter (cp) is the value that can be tuned here to determine the optimal minimum by which adding another split must improve the model (Irizarry, n.d.b). Using `rpart`, I developed a classification tree based on the available factors, using different complexity parameters. The model determined what factors and what cutoff points should be used to classify the messages.

```
class_tree_fit <- train(
  type ~ .,
  method = "rpart",
  data=train_set_features,
  trControl = control,
  metric = "F1",
  tuneGrid = data.frame(cp = seq(0.0, 0.1, len = 25)),
)
class_tree_y_hat <- predict(class_tree_fit, newdata = test_set, type="raw")
class_tree_f_meas <- F_meas(
  class_tree_y_hat,
  reference = test_set$type,
  relevant = "spam",
  beta = 0.75
)
class_tree_f_meas
```

```
## [1] 0.9563212
```

Plotting the final model shows what variables and cut-off values were used.

```
plot(class_tree_fit$finalModel, margin = 0.1, main="Final classification tree")
text(class_tree_fit$finalModel, cex = 0.75)
```



The model only needed the number of occurrences of top spam words and the length of the longest number to predict the outcome.

Model 6: Random Forest A random forest builds on the classification tree idea by randomly selecting bootstrap samples and building classification trees from the samples, using randomly selected features. The resulting trees create the "forest", and the average prediction of all the trees determines the final prediction (Irizarry, n.d.b). While the classification tree model did quite well with only two features, a random forest model would potentially look at more features and might therefore be more flexible.

```
random_forest_fit <- train(
  type ~ .,
  method = "rf",
  data = train_set_features,
  trControl = control,
  metric = "F1",
  # Tune the number of variables randomly assigned per split
  tuneGrid=data.frame(mtry=seq(1, 6))
)
random_forest_y_hat <- predict(random_forest_fit, newdata = test_set, type="raw")
random_forest_f_meas <- F_meas(
  random_forest_y_hat,
  reference = test_set$type,
  relevant = "spam",
  beta = 0.75
)
random_forest_f_meas
```

```
## [1] 0.968939
```

When reviewing the relative importance of the various factors, I could see that `top_spam_words_count` had the greatest impact. That was the same factor that was used in the first split in the classification tree model.

	MeanDecreaseGini
top_spam_words_count	279.86764
num_count	206.84614
longest_number	228.28684
capital_count	91.57365
longest_capitals	100.09202
msg_length	65.66871

While the random forest model did successfully incorporate other features, the presence or absence of words in the most-common spam words list seems to be a strong indicator a message is spam.

Ensembles

After testing models one at a time, I considered whether combining predictions from multiple models could yield better results.

One challenge was that I wanted to only use models that were among the most accurate, but I could not use the previously computed accuracy results, because they were based on test data. Using the results from testing on the test set to determine which model to use would pollute my data.

Instead, I used the same bootstrap approach used for tuning the model parameters. I created five bootstrap samples from the training data, predicted the outcome for each using the existing models, and averaged the resulting F-values.

Models	F-Values
Guessing	0.1366253
Longest number	0.9200773
Logistic regression	0.9147975
K-Nearest Neighbor	0.9510772
Classification Tree	0.9696846
Random Forest	0.9825162

Five of the models had F-values of greater than 0.9 against the training data bootstrap samples.

Voting The first ensemble model I tried was a voting method. Since Longest number and Logistic regression were both models based on the `longest_number` feature, I only used one of them. I chose Longest number because it performed better in my bootstrap sample.

This left me with an even number of models, so I had to determine how to handle ties. Since I had decided that not misidentifying ham was more important than not misidentifying spam, in case of a tie, the message was classified as ham. That meant that 3 out of the 4 models had to consider a message spam in order to classify it as such.

```r
ensemble_vote_fit <- cbind(long_number_y_hat,
                           class_tree_y_hat,
                           random_forest_y_hat,
                           knn_y_hat)
ensemble_vote_y_hat <- apply(ensemble_vote_fit, 1, function(r) {
  ifelse(sum(r == 2) >= 3, 'spam', 'ham')
```

```r
}) |> factor(levels = levels(test_set$type))
ensemble_vote_f_meas <- F_meas(
  ensemble_vote_y_hat,
  reference = test_set$type,
  relevant = "spam",
  beta = 0.75
)
ensemble_vote_f_meas
```

```
## [1] 0.9431225
```

Mean Probability The second ensemble model I tried was by calculating the probability of a message being spam, using the top three models: KNN, Classification Tree, and Random Forest. I took the mean of the three probabilities and used that mean probability to predict whether the message was spam or ham.

```r
# Re-generate predictions to get the probability of the item being spam
knn_p_hat <- predict(
  knn_fit,
  newdata = test_set ,
  type = "prob"
  )['spam']
class_tree_p_hat <- predict(
  class_tree_fit,
  newdata = test_set,
  type = "prob"
  )['spam']
random_forest_p_hat <- predict(
  random_forest_fit,
  newdata = test_set,
  type = "prob"
  )['spam']
ensemble_mean_fit <- cbind(knn_p_hat, class_tree_p_hat, random_forest_p_hat)
# Get the mean probability across all three models. If it's greater than 0.5
# likelihood of being spam, classify it as such
ensemble_mean_y_hat <- apply(ensemble_mean_fit, 1, function(r) {
  ifelse(mean(r) > 0.5, 'spam', 'ham')
}) |> factor(levels = levels(test_set$type))
ensemble_mean_f_meas <- F_meas(
  ensemble_mean_y_hat,
  reference = test_set$type,
  relevant = "spam",
  beta = 0.75
)
ensemble_mean_f_meas
```

```
## [1] 0.966317
```

Results

While the ensemble mean approach improved the results over the classification tree and KNN models on their own, the random forest model gave the most accurate results overall.

Table 8: Model performance on test data set

Model	F1 Score
Guessing	0.1625065
Longest number	0.9226714
Logistic regression	0.8972943
K-Nearest Neighbor	0.9226714
Classification Tree	0.9563212
Random Forest	0.9689390
Ensemble Voting	0.9431225
Ensemble Mean	0.9663170

The full confusion matrix details that the random forest model has specificity/recall over 99% and a slightly lower sensitivity/precision. It correctly classified 99% of the messages in the test set overall and 94% of the spam messages. Importantly, only 2 of the 966 ham messages were incorrectly classified as spam.

```r
confusionMatrix(random_forest_y_hat, test_set$type, positive = "spam")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction ham spam
##       ham  964    9
##       spam   2  141
##
##                Accuracy : 0.9901
##                  95% CI : (0.9824, 0.9951)
##     No Information Rate : 0.8656
##     P-Value [Acc > NIR] : < 2e-16
##
##                   Kappa : 0.9568
##
##  Mcnemar's Test P-Value : 0.07044
##
##             Sensitivity : 0.9400
##             Specificity : 0.9979
##          Pos Pred Value : 0.9860
##          Neg Pred Value : 0.9908
##              Prevalence : 0.1344
##          Detection Rate : 0.1263
##    Detection Prevalence : 0.1281
##       Balanced Accuracy : 0.9690
##
##        'Positive' Class : spam
##
```

Conclusion

Overall, a random forest model works well for classifying spam messages using factors such as the presence of words known to be used more often in spam messages, long strings of digits such as phone numbers, and numbers of digits or capital letters.

Limitations and Further Research

The spam detection model developed is limited by the dataset. While the dataset was, at time of release, the largest available corpus of SMS spam messages (Almeida, María Gómez Hidalgo, and Yamakami 2011), five thousand messages represent only the tinest fraction of the trillions of text messages sent today.

In addition, the dataset was released in 2011. The past fifteen years have seen transformations in communication, with the rise of smart phones, emojis, and new messaging apps. Thus, further research is needed to determine if the model developed here performs as well with a more recent dataset.

Finally, the dataset and models use only the text of the message in classification. Further research is needed to determine whether context outside of the message text could more accurately indicate whether a message is spam. For instance, the technology used to send the message or whether the sender and recipient have communicated before could help classify a message.

Another limitation and challenge of designing any sort of spam classification is that there is no one single standard for what is a spam message. A frequent grey area is messages from political campaigns. While they do not meet the CTIA definition of an "Unwanted message" (spam) in most cases (CTIA 2023), many people who receive them consider them no less annoying than an unsolicited ad. My work here simply accepts the classification of "spam" or "ham" in the data set at face value.

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